

The circuit for stepper motor connections

The circuit diagram for controlling the unipolar stepper motor is shown in Fig. 2. The `camstepper.php` program will control the stepper motor. The unipolar stepper has five or six wires—the middle or common wires are joined together and connected to +5V supply. The remaining four wires are connected to pins 13, 14, 15 and 16 of ULN2003.

For a 5V bipolar stepper, connect it directly to the GPIOs, bypassing the ULN2003. That way it produces less but good enough torque for a tiny webcam. However, for better torque, it is to be connected through a ULN2003.

If you study the `camstepper.php` program, you would find that the entire pan of the webcam is controlled by just two GPIOs. These two GPIOs are GPIO-18 and GPIO-21 (physical pins 12 and 13). Any other GPIO can also be considered. The two GPIOs have the following states:

```
0 0 ---- No motion
1 0 ---- Clockwise motion of webcam
0 1 ---- Anti-clockwise motion of webcam
1 1 ---- Oscillating motion of webcam.
```

The `gpio.php` code given below shows how to achieve the same:

```
$sudo nano /var/www/gpio.php
<?php
include ("http_auth.php"); //this is a
login - password window
?>
<html>
<head>
<title>Bera world webcam tracking control
center</title>
</head>
<body>
</body>
<form action="gpio.php" method="get">
<input type="submit" value="Forward On">
<input type="hidden" name="fwd"
value="on">
</form>
<form action="gpio.php" method="get">
<input type="submit" value="Forward Off">
<input type="hidden" name="fwd"
value="off">
</form>
<form action="gpio.php" method="get">
```

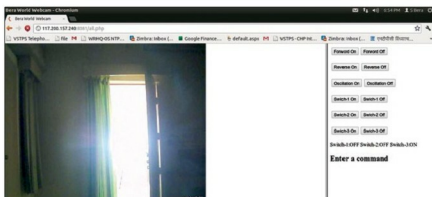


Fig. 3: Combined browser window for image being captured and stepper motor control panel

```
<input type="submit" value="Reverse On">
<input type="hidden" name="rev"
value="on">
</form>
<form action="gpio.php" method="get">
<input type="submit" value="Reverse Off">
<input type="hidden" name="rev"
value="off">
</form>
<form action="gpio.php" method="get">
<input type="submit" value="Oscillation
On">
<input type="hidden" name="osc"
value="on">
</form>
<form action="gpio.php" method="get">
<input type="submit" value="Oscillation
Off">
<input type="hidden" name="osc"
value="off">
</form>
</html>

<?php
If (isset($_GET['fwd'])) {
    If ($_GET['fwd'] == "on") {
        $page = shell_exec('gpio -g mode 18
out');
        $page = shell_exec('gpio -g write 18 1');
        $page = shell_exec('gpio -g mode 21
out');
        $page = shell_exec('gpio -g write 21 1');
    } else If ($_GET['osc'] == "off") {
        $page = shell_exec('gpio -g write 18 0');
        $page = shell_exec('gpio -g write 21 0');
    }
}

$al = shell_exec('gpio -g read 18'); //
read the status of GPIO-18
$b1 = shell_exec('gpio -g read 21'); //
read the status of GPIO-21

if ($al==0 && $b1==0) echo "<h2> Enter a
command";
if ($al==1 && $b1==0) echo "<h2> Forward
motion is on";
if ($al==0 && $b1==1) echo "<h2> Reverse
motion is on";
if ($al==1 && $b1==1) echo "<h2>
Oscillation is on";
?>

At the top, we included an http_
auth.php program only to safeguard
against getting the webcam manipu-
lated by every Tom, Dick and Harry. It
is a login-password program.

$sudo nano http_auth.php
<?php
$valid_passwords = array ("pi-pepper" =>
"pi123", "beracan" => "beral23");
$valid_users = array_keys($valid_
passwords);
```